In [1]:

```
# ============================================================
# EXERCISE 1 : DATA PREPROCESSING AND CLEANING
# ============================================================

# 1. Import Required Libraries

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.preprocessing import LabelEncoder, StandardScaler


# ============================================================
# 2. Load Dataset
# ============================================================

df = pd.read_csv("C:/Users/Lenovo/Downloads/archive (1)/Telco_Cusomer_Churn.csv")

print("FIRST 10 RECORDS")
print(df.head(10))

print("\nLAST 10 RECORDS")
print(df.tail(10))

print("\nSHAPE OF DATASET")
print(df.shape)

print("\nCOLUMN NAMES")
print(df.columns.tolist())


# ============================================================
# Convert TotalCharges to Numeric
# ============================================================

if 'TotalCharges' in df.columns:
    df['TotalCharges'] = pd.to_numeric(df['TotalCharges'], errors='coerce')


# ============================================================
# 3. Explore Dataset
# ============================================================

print("\nDATASET INFORMATION")
print(df.info())

print("\nSUMMARY STATISTICS")
print(df.describe(include='all'))

print("\nMISSING VALUES")
print(df.isnull().sum())

print("\nDUPLICATE RECORDS")
print(df.duplicated().sum())
```

```
# ============================================================
# 4(a). Remove Duplicate Records
# ============================================================

df.drop_duplicates(inplace=True)

print("\nDuplicates After Removal")
print(df.duplicated().sum())


# ============================================================
# 4(b). Replace Missing Values
# ============================================================

# Numerical Columns

num_cols = df.select_dtypes(include=['int64', 'float64']).columns

for col in num_cols:
    df[col] = df[col].fillna(df[col].median())

# Categorical Columns

cat_cols = df.select_dtypes(include=['object']).columns

for col in cat_cols:
    df[col] = df[col].fillna(df[col].mode()[0])


# ============================================================
# 4(c). Verify Missing Values
# ============================================================

print("\nMissing Values After Cleaning")
print(df.isnull().sum())


# ============================================================
# 5. Encode Categorical Columns
# ============================================================

# Binary Columns

binary_cols = ['gender','Partner','Dependents','PhoneService',
               'PaperlessBilling','Churn']

le = LabelEncoder()

for col in binary_cols:
    if col in df.columns:
        df[col] = le.fit_transform(df[col])

# One-Hot Encoding

multi_cols = ['Contract','PaymentMethod','InternetService']

existing_cols = [c for c in multi_cols if c in df.columns]

df = pd.get_dummies(df,
                    columns=existing_cols,
```

```
                    drop_first=True)

print("\nDataset After Encoding")
print(df.head())


# ============================================================
# 6. Standardize Numerical Features
# ============================================================

scaler = StandardScaler()

scale_cols = []

for col in ['tenure','MonthlyCharges','TotalCharges']:
    if col in df.columns:
        scale_cols.append(col)

df[scale_cols] = scaler.fit_transform(df[scale_cols])

print("\nScaled Features")
print(df[scale_cols].head())


# ============================================================
# 7. Distribution of MonthlyCharges
# ============================================================

if 'MonthlyCharges' in df.columns:

    plt.figure(figsize=(7,5))

    sns.histplot(df['MonthlyCharges'],
                 kde=True,
                 bins=30)

    plt.title("Distribution of Monthly Charges")
    plt.show()


# ============================================================
# Distribution of Tenure
# ============================================================

if 'tenure' in df.columns:

    plt.figure(figsize=(7,5))

    sns.histplot(df['tenure'],
                 kde=True,
                 bins=30)

    plt.title("Distribution of Tenure")
    plt.show()


# ============================================================
# 8. Count Plot
# Contract vs Churn
# ============================================================
```

```
if 'Contract' in df.columns:

    plt.figure(figsize=(8,5))

    sns.countplot(x='Contract',
                  hue='Churn',
                  data=df)

    plt.title("Contract vs Churn")
    plt.xticks(rotation=15)
    plt.show()

else:

    contract_cols = [c for c in df.columns if c.startswith("Contract_")]

    if len(contract_cols) > 0:

        plt.figure(figsize=(8,5))

        sns.countplot(x=contract_cols[0],
                      hue='Churn',
                      data=df)

        plt.title("Contract vs Churn")
        plt.show()


# ============================================================
# 9. Box Plot
# ============================================================

if 'MonthlyCharges' in df.columns:

    plt.figure(figsize=(7,5))

    sns.boxplot(x='Churn',
                y='MonthlyCharges',
                data=df)

    plt.title("Monthly Charges by Churn")
    plt.show()


# ============================================================
# 10. Correlation Heatmap
# ============================================================

numeric_df = df.select_dtypes(include=['int64','float64'])

plt.figure(figsize=(15,10))

sns.heatmap(numeric_df.corr(),
            annot=True,
            cmap='coolwarm',
            fmt=".2f")

plt.title("Correlation Heatmap")
plt.show()
```

FIRST 10 RECORDS

	customerID	gender	SeniorCitizen	Partner	Dependents	tenure	PhoneService	\
0	7590-VHVEG	Female	0	Yes	No	1	No	
1	5575-GNVDE	Male	0	No	No	34	Yes	
2	3668-QPYBK	Male	0	No	No	2	Yes	
3	7795-CFOCW	Male	0	No	No	45	No	
4	9237-HQITU	Female	0	No	No	2	Yes	
5	9305-CDSKC	Female	0	No	No	8	Yes	
6	1452-KIOVK	Male	0	No	Yes	22	Yes	
7	6713-OKOMC	Female	0	No	No	10	No	
8	7892-POOKP	Female	0	Yes	No	28	Yes	
9	6388-TABGU	Male	0	No	Yes	62	Yes	

	MultipleLines	InternetService	OnlineSecurity	...	DeviceProtection	\
0	No phone service	DSL	No	...	No	
1	No	DSL	Yes	...	Yes	
2	No	DSL	Yes	...	No	
3	No phone service	DSL	Yes	...	Yes	
4	No	Fiber optic	No	...	No	
5	Yes	Fiber optic	No	...	Yes	
6	Yes	Fiber optic	No	...	No	
7	No phone service	DSL	Yes	...	No	
8	Yes	Fiber optic	No	...	Yes	
9	No	DSL	Yes	...	No	

	TechSupport	StreamingTV	StreamingMovies	Contract	PaperlessBilling	\
0	No	No	No	Month-to-month	Yes	
1	No	No	No	One year	No	
2	No	No	No	Month-to-month	Yes	
3	Yes	No	No	One year	No	
4	No	No	No	Month-to-month	Yes	
5	No	Yes	Yes	Month-to-month	Yes	
6	No	Yes	No	Month-to-month	Yes	
7	No	No	No	Month-to-month	No	
8	Yes	Yes	Yes	Month-to-month	Yes	
9	No	No	No	One year	No	

	PaymentMethod	MonthlyCharges	TotalCharges	Churn
0	Electronic check	29.85	29.85	No
1	Mailed check	56.95	1889.5	No
2	Mailed check	53.85	108.15	Yes
3	Bank transfer (automatic)	42.30	1840.75	No
4	Electronic check	70.70	151.65	Yes
5	Electronic check	99.65	820.5	Yes
6	Credit card (automatic)	89.10	1949.4	No
7	Mailed check	29.75	301.9	No
8	Electronic check	104.80	3046.05	Yes
9	Bank transfer (automatic)	56.15	3487.95	No

[10 rows x 21 columns]

LAST 10 RECORDS

	customerID	gender	SeniorCitizen	Partner	Dependents	tenure	\
7033	9767-FFLEM	Male	0	No	No	38	
7034	0639-TSIQW	Female	0	No	No	67	
7035	8456-QDAVC	Male	0	No	No	19	
7036	7750-EYXWZ	Female	0	No	No	12	
7037	2569-WGERO	Female	0	No	No	72	
7038	6840-RESVB	Male	0	Yes	Yes	24	
7039	2234-XADUH	Female	0	Yes	Yes	72	

```
7040  4801-JZAZL  Female              0     Yes        Yes      11
7041  8361-LTMKD    Male              1     Yes         No       4
7042  3186-AJIEK    Male              0      No         No      66

     PhoneService     MultipleLines InternetService       OnlineSecurity  ...  \
7033          Yes                No     Fiber optic                   No  ...
7034          Yes               Yes     Fiber optic                  Yes  ...
7035          Yes                No     Fiber optic                   No  ...
7036           No  No phone service             DSL                   No  ...
7037          Yes                No              No  No internet service  ...
7038          Yes               Yes             DSL                  Yes  ...
7039          Yes               Yes     Fiber optic                   No  ...
7040           No  No phone service             DSL                  Yes  ...
7041          Yes               Yes     Fiber optic                   No  ...
7042          Yes                No     Fiber optic                  Yes  ...

        DeviceProtection          TechSupport          StreamingTV  \
7033                  No                   No                  No
7034                 Yes                   No                 Yes
7035                  No                   No                 Yes
7036                 Yes                  Yes                 Yes
7037  No internet service  No internet service  No internet service
7038                 Yes                  Yes                 Yes
7039                 Yes                   No                 Yes
7040                  No                   No                  No
7041                  No                   No                  No
7042                 Yes                  Yes                 Yes

         StreamingMovies        Contract PaperlessBilling  \
7033                  No  Month-to-month              Yes
7034                  No  Month-to-month              Yes
7035                  No  Month-to-month              Yes
7036                 Yes        One year               No
7037  No internet service        Two year              Yes
7038                 Yes        One year              Yes
7039                 Yes        One year              Yes
7040                  No  Month-to-month              Yes
7041                  No  Month-to-month              Yes
7042                 Yes        Two year              Yes

                 PaymentMethod MonthlyCharges  TotalCharges Churn
7033    Credit card (automatic)          69.50       2625.25    No
7034    Credit card (automatic)         102.95       6886.25   Yes
7035  Bank transfer (automatic)          78.70        1495.1    No
7036           Electronic check          60.65         743.3    No
7037  Bank transfer (automatic)          21.15        1419.4    No
7038               Mailed check          84.80        1990.5    No
7039    Credit card (automatic)         103.20        7362.9    No
7040           Electronic check          29.60        346.45    No
7041               Mailed check          74.40         306.6   Yes
7042  Bank transfer (automatic)         105.65        6844.5    No

[10 rows x 21 columns]

SHAPE OF DATASET
(7043, 21)

COLUMN NAMES
['customerID', 'gender', 'SeniorCitizen', 'Partner', 'Dependents', 'tenure', 'PhoneSe
rvice', 'MultipleLines', 'InternetService', 'OnlineSecurity', 'OnlineBackup', 'Device
```

```
Protection', 'TechSupport', 'StreamingTV', 'StreamingMovies', 'Contract', 'PaperlessB
illing', 'PaymentMethod', 'MonthlyCharges', 'TotalCharges', 'Churn']

DATASET INFORMATION
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 7043 entries, 0 to 7042
Data columns (total 21 columns):
 #   Column            Non-Null Count  Dtype
---  ------            --------------  -----
 0   customerID        7043 non-null   object
 1   gender            7043 non-null   object
 2   SeniorCitizen     7043 non-null   int64
 3   Partner           7043 non-null   object
 4   Dependents        7043 non-null   object
 5   tenure            7043 non-null   int64
 6   PhoneService      7043 non-null   object
 7   MultipleLines     7043 non-null   object
 8   InternetService   7043 non-null   object
 9   OnlineSecurity    7043 non-null   object
 10  OnlineBackup      7043 non-null   object
 11  DeviceProtection  7043 non-null   object
 12  TechSupport       7043 non-null   object
 13  StreamingTV       7043 non-null   object
 14  StreamingMovies   7043 non-null   object
 15  Contract          7043 non-null   object
 16  PaperlessBilling  7043 non-null   object
 17  PaymentMethod     7043 non-null   object
 18  MonthlyCharges    7043 non-null   float64
 19  TotalCharges      7032 non-null   float64
 20  Churn             7043 non-null   object
dtypes: float64(2), int64(2), object(17)
memory usage: 1.1+ MB
None

SUMMARY STATISTICS
        customerID gender  SeniorCitizen Partner Dependents       tenure  \
count         7043   7043    7043.000000    7043       7043  7043.000000
unique        7043      2            NaN       2          2          NaN
top     7590-VHVEG   Male            NaN      No         No          NaN
freq             1   3555            NaN    3641       4933          NaN
mean           NaN    NaN       0.162147     NaN        NaN    32.371149
std            NaN    NaN       0.368612     NaN        NaN    24.559481
min            NaN    NaN       0.000000     NaN        NaN     0.000000
25%            NaN    NaN       0.000000     NaN        NaN     9.000000
50%            NaN    NaN       0.000000     NaN        NaN    29.000000
75%            NaN    NaN       0.000000     NaN        NaN    55.000000
max            NaN    NaN       1.000000     NaN        NaN    72.000000

       PhoneService MultipleLines InternetService OnlineSecurity  ...  \
count          7043          7043            7043           7043  ...
unique            2             3               3              3  ...
top             Yes            No     Fiber optic             No  ...
freq           6361          3390            3096           3498  ...
mean            NaN           NaN             NaN            NaN  ...
std             NaN           NaN             NaN            NaN  ...
min             NaN           NaN             NaN            NaN  ...
25%             NaN           NaN             NaN            NaN  ...
50%             NaN           NaN             NaN            NaN  ...
75%             NaN           NaN             NaN            NaN  ...
max             NaN           NaN             NaN            NaN  ...
```

	DeviceProtection	TechSupport	StreamingTV	StreamingMovies	\
count	7043	7043	7043	7043	
unique	3	3	3	3	
top	No	No	No	No	
freq	3095	3473	2810	2785	
mean	NaN	NaN	NaN	NaN	
std	NaN	NaN	NaN	NaN	
min	NaN	NaN	NaN	NaN	
25%	NaN	NaN	NaN	NaN	
50%	NaN	NaN	NaN	NaN	
75%	NaN	NaN	NaN	NaN	
max	NaN	NaN	NaN	NaN	

	Contract	PaperlessBilling	PaymentMethod	MonthlyCharges	\
count	7043	7043	7043	7043.000000	
unique	3	2	4	NaN	
top	Month-to-month	Yes	Electronic check	NaN	
freq	3875	4171	2365	NaN	
mean	NaN	NaN	NaN	64.761692	
std	NaN	NaN	NaN	30.090047	
min	NaN	NaN	NaN	18.250000	
25%	NaN	NaN	NaN	35.500000	
50%	NaN	NaN	NaN	70.350000	
75%	NaN	NaN	NaN	89.850000	
max	NaN	NaN	NaN	118.750000	

	TotalCharges	Churn
count	7032.000000	7043
unique	NaN	2
top	NaN	No
freq	NaN	5174
mean	2283.300441	NaN
std	2266.771362	NaN
min	18.800000	NaN
25%	401.450000	NaN
50%	1397.475000	NaN
75%	3794.737500	NaN
max	8684.800000	NaN

[11 rows x 21 columns]

MISSING VALUES

customerID	0
gender	0
SeniorCitizen	0
Partner	0
Dependents	0
tenure	0
PhoneService	0
MultipleLines	0
InternetService	0
OnlineSecurity	0
OnlineBackup	0
DeviceProtection	0
TechSupport	0
StreamingTV	0
StreamingMovies	0
Contract	0
PaperlessBilling	0

```
PaymentMethod        0
MonthlyCharges       0
TotalCharges        11
Churn                0
dtype: int64

DUPLICATE RECORDS
0

Duplicates After Removal
0

Missing Values After Cleaning
customerID          0
gender              0
SeniorCitizen       0
Partner             0
Dependents          0
tenure              0
PhoneService        0
MultipleLines       0
InternetService     0
OnlineSecurity      0
OnlineBackup        0
DeviceProtection    0
TechSupport         0
StreamingTV         0
StreamingMovies     0
Contract            0
PaperlessBilling    0
PaymentMethod       0
MonthlyCharges      0
TotalCharges        0
Churn               0
dtype: int64

Dataset After Encoding
   customerID  gender  SeniorCitizen  Partner  Dependents  tenure  \
0  7590-VHVEG       0              0        1           0       1
1  5575-GNVDE       1              0        0           0      34
2  3668-QPYBK       1              0        0           0       2
3  7795-CFOCW       1              0        0           0      45
4  9237-HQITU       0              0        0           0       2

   PhoneService     MultipleLines OnlineSecurity OnlineBackup  ...  \
0             0  No phone service             No          Yes  ...
1             1                No            Yes           No  ...
2             1                No            Yes          Yes  ...
3             0  No phone service            Yes           No  ...
4             1                No             No           No  ...

  MonthlyCharges TotalCharges Churn Contract_One year  Contract_Two year  \
0          29.85        29.85     0                 0                  0
1          56.95      1889.50     0                 1                  0
2          53.85       108.15     1                 0                  0
3          42.30      1840.75     0                 1                  0
4          70.70       151.65     1                 0                  0

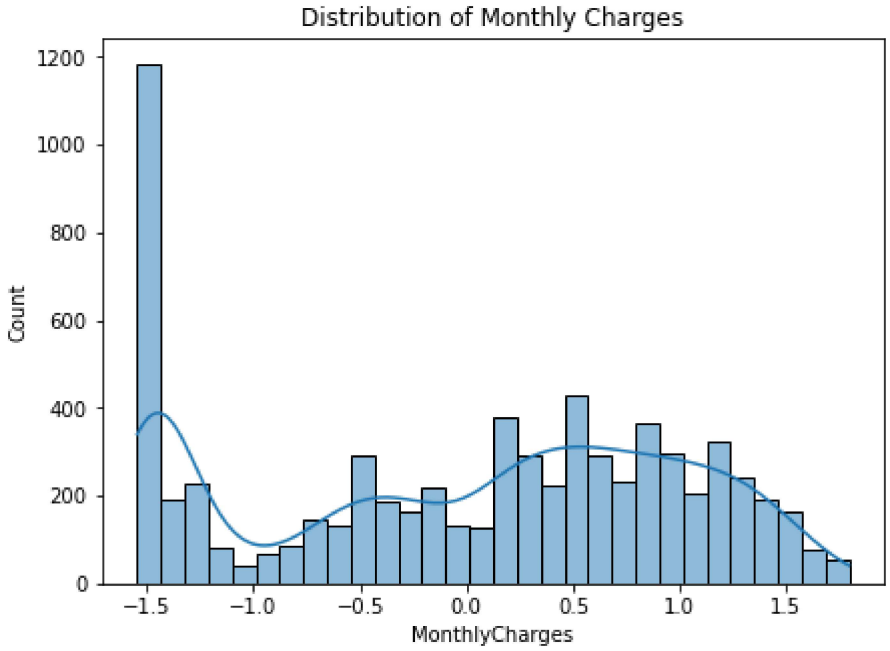
   PaymentMethod_Credit card (automatic)  PaymentMethod_Electronic check  \
0                                      0                               1
```

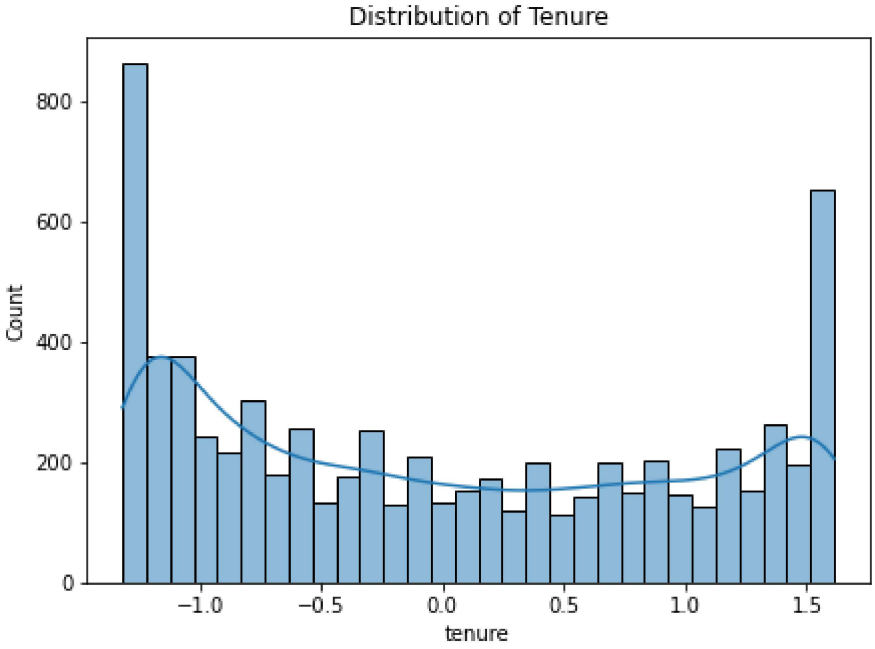
```
1                                    0                                    0
2                                    0                                    0
3                                    0                                    0
4                                    0                                    1

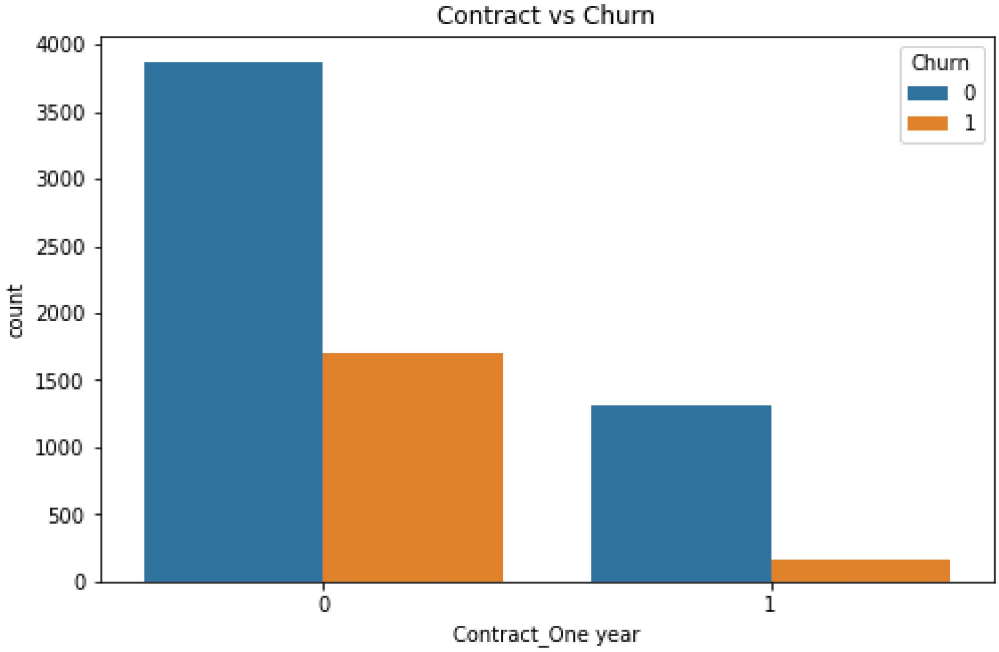
   PaymentMethod_Mailed check  InternetService_Fiber optic  InternetService_No
0                           0                             0                   0
1                           1                             0                   0
2                           1                             0                   0
3                           0                             0                   0
4                           0                             1                   0

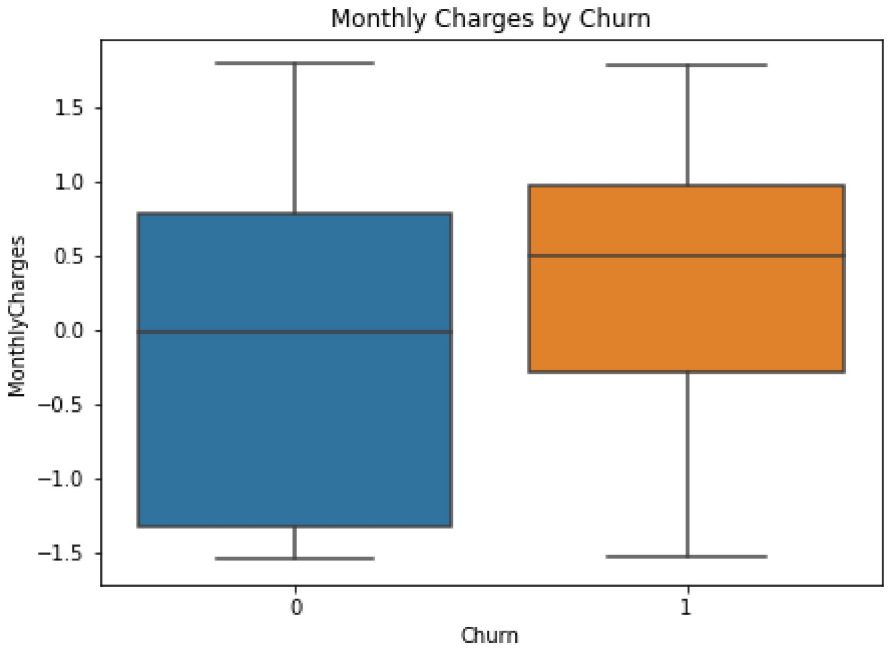
[5 rows x 25 columns]

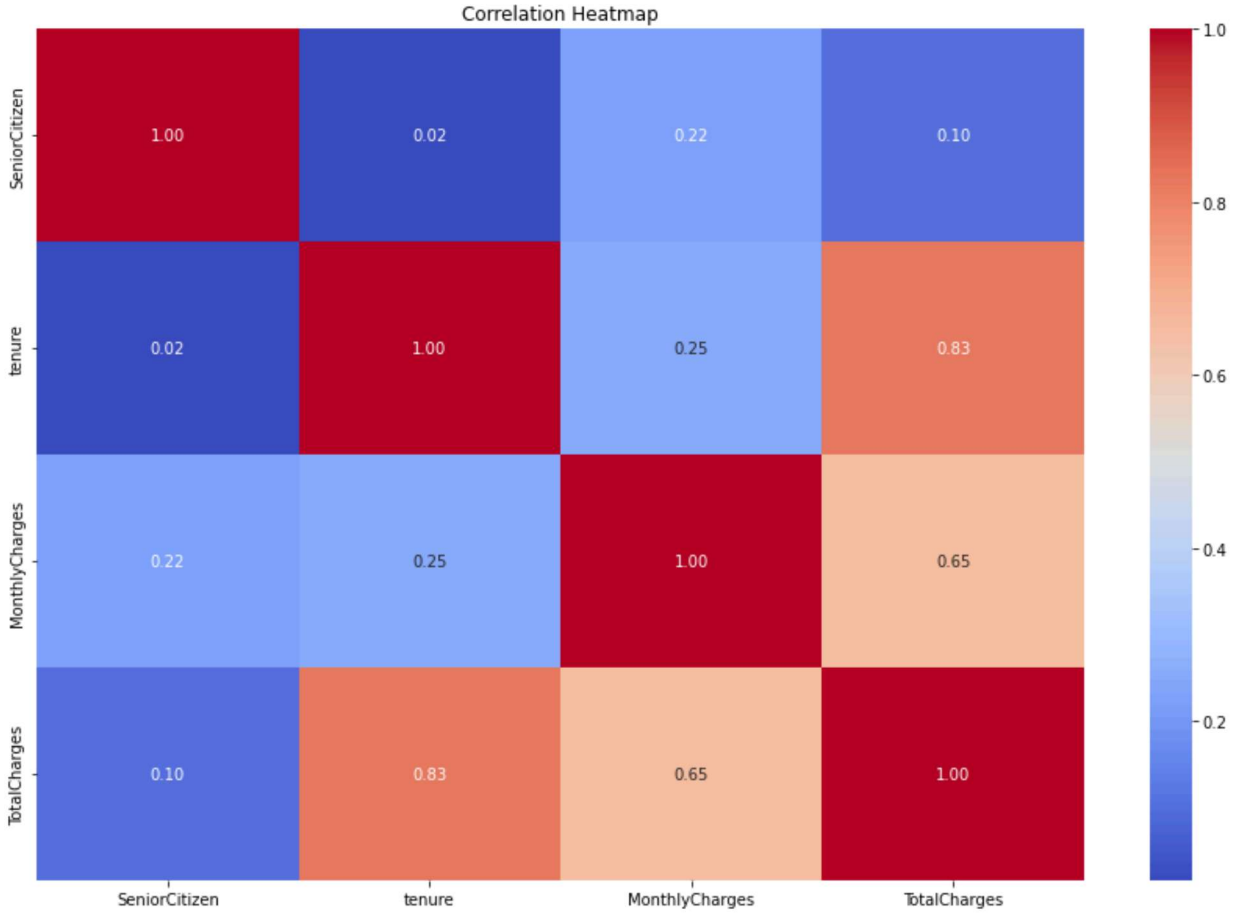
Scaled Features
     tenure  MonthlyCharges  TotalCharges
0 -1.277445       -1.160323     -0.994242
1  0.066327       -0.259629     -0.173244
2 -1.236724       -0.362660     -0.959674
3  0.514251       -0.746535     -0.194766
4 -1.236724        0.197365     -0.940470
```




Distribution of Tenure
Count
tenure
800
600
400
200
0
−1.0
−0.5
0.0
0.5
1.0
1.5


Contract vs Churn
Churn
0
1
count
Contract_One year
4000
3500
3000
2500
2000
1500
1000
500
0
0
1


Monthly Charges by Churn
MonthlyCharges
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
0
1
Churn


Correlation Heatmap
SeniorCitizen
tenure
MonthlyCharges
TotalCharges
1.00
0.02
0.22
0.10
0.02
1.00
0.25
0.83
0.22
0.25
1.00
0.65
0.10
0.83
0.65
1.00
SeniorCitizen
tenure
MonthlyCharges
TotalCharges
1.0
0.8
0.6
0.4
0.2

In [ ]: